

Multiple Disease Prediction from Retinal Images Using Deep Learning Algorithm

* B. Rishikesh¹

PG Student,

Department of Computer Science and Engineering,
SRM Institute of Science and Technology, Vadapalani Campus,
Chennai-26, Tamilnadu, India.
rishikesh494@gmail.com

K. Karthikayani²

Assistant Professor,

Department of Computer Science and Engineering,
SRM Institute of Science and Technology, Vadapalani Campus,
Chennai-26, Tamilnadu, India.
karthikk3@srmist.edu.in

Abstract- Retinal imaging serves as an essential tool for the early identification and diagnosis of numerous systemic diseases. This model framework is used for predicting multiple diseases simultaneously using retinal images. The implementation begins with the collection of annotated retinal images, enabling the identification of ailments as glaucoma, age-related macular degeneration, diabetic retinopathy, including hypertensive retinopathy. Preprocessing techniques are applied to enhance data quality and reduce noise. Feature extraction is conducted using the pretrained VGG16 convolutional neural network (CNN) essential for deep learning model training. Multi-label categorization and multi-task learning are employed to predict multiple diseases concurrently. Strict measures, including accuracy, sensitivity, specificity, including the area under the receiver operating characteristic curve (AUC-ROC), are used to assess the system's performance. Clinical validation also evaluates its efficacy in practical settings. The incorporation of this predictive system into clinical workflows is facilitated by ensuring regulatory compliance and seamless communication with healthcare professionals. Experimental Results show that the suggested VGG16-based system accomplishes an impressive 98.7% accuracy in detecting multiple retinal diseases. The implementation underscores the transformative potential of leveraging advanced deep learning techniques in ophthalmology. Future efforts are focused on enhancing algorithm performance through larger datasets and refining models to further improve diagnostic precision, enabling timely and effective treatment strategies for retinal conditions.

Keywords– Retinal Imaging, Deep Learning, VGG16, Disease Prediction, Multi-label Classification, Retinal Disorders.

I. INTRODUCTION

To perceive our surroundings, the human retina must convert light into visual data. The retina is vulnerable to numerous disorders/conditions it, not identified and treated promptly, could result in blindness or irreversible vision loss, including diabetic retinopathy, glaucoma, age-related macular degeneration, and retinal detachment. Early on detection of these retinal conditions is crucial for effective management and improved patient outcomes. Convolutional neural networks, or CNNs, are a major development in artificial intelligence and deep learning that have significantly improved automated diagnostic instruments in ophthalmology. Known for their exceptional performance in image processing, CNNs have proven highly effective in identifying and classifying retinal diseases. This study aims to leverage CNNs to predict and diagnose different retinal disorders using digitized retinal images. We seek to train a CNN model with a comprehensive and well-annotated retinal image dataset to accurately detect the presence and severity of various retinal conditions. The use of CNNs offers several benefits, including the capability to detect intricate patterns

and features in retinal images that may go unnoticed by human observers, as well as speeding up the diagnostic process for timely interventions. Moreover, the application of automated systems in areas with limited access to specialists can help bridge the gap in the availability of qualified ophthalmologists. The structure of this document is as follows: Relevant literature is examined in Section II, and Section III outlines current systems, Section IV elaborates on the proposed system, Section V emphasizes results and discussions, and Section VI presents the conclusion.

II. RELATED WORK

This study introduces the RAVIR dataset, developed to use infrared reflectance imaging in order to segment retinal veins and arteries semantically. This dataset enables the development of deep learning models that can categorize blood vessels with minimal post-processing requirements. We provide SegRAVIR, an advanced deep learning framework that executes quantitative evaluation along with semantic segmentation of retinal veins and arteries, facilitating accurate measurement of vessel diameters. Extensive research demonstrates that SegRAVIR outperforms existing models, validating its effectiveness in retinal vessel segmentation and analysis. Furthermore, we present a knowledge distillation method to modify RAVIR-pretrained networks for application with color images. SegRAVIR facilitates complete automation of segmentation and analysis of retinal vasculature, eliminating the need for supplementary preprocessing or post-processing. It precisely quantifies vessel sizes, essential for quantitative retinal evaluations. Our methodology exhibits significant generalizability by surpassing prior techniques on the RITE dataset for artery and vein segmentation. Moreover, we demonstrate that knowledge distillation markedly improves domain adaptability for retinal vasculature segmentation across several imaging modalities. Our dataset and model offer a comprehensive foundation for sophisticated retinal analysis employing diverse imaging modalities[1].

This study tackles a significant difficulty in the precise visualization, diagnosis, treatment, and surgical planning for ocular conditions: retinal vascular segmentation. Recent deep learning methodologies have attained significant outcomes in this field; nonetheless, they frequently depend on global preprocessing and train on the complete retinal image. This method has two major limitations: the failure to utilize localized patch information and the disregard for geometric constraints, as the retina is restricted to particular areas within the retrieved patches or the entire image. As a result, these approaches encounter difficulties in identifying minute, slender arteries and differentiating the optic disc. [2]

We present the Global along with Local Enhanced Residual U-Net (GLUE) to address these constraints and

achieve accurate retinal vascular segmentation. This model amalgamates localized and global data within the retinal area to enhance segmentation precision. Experimental findings on two benchmark datasets indicate that GLUE routinely surpasses state-of-the-art methodologies and attains competitive outcomes relative to conventional U-Net structures. Efficient retinal vessel segmentation is essential for identifying problems such as diabetic retinopathy, cardiovascular disease, and chronic ocular disorders. Nonetheless, obstacles such as inadequate lighting, absent minor vessels, and intricate vessel morphology complicate this endeavor. Prior learning-based models utilized image-level pre-processing on whole retinal pictures, frequently including extraneous pixel data from beyond the retina. This study integrates CLAHE activities locally and globally to enhance segmentation performance. The GLUE model combines a Weighted U-Net (WUN) with coarse segmentation and a Weighted Residual U-Net (WRUN) over fine-tuning segmentation effectively addressing the complexities of retinal vascular segmentation[3].

This research presents a an A/V discriminator-based multi-scale interactive network to enhance the identification of retinal veins and arteries, tackling issues related to noisy labeling and arteriovenous ambiguity. An MI module within the encoder facilitates cross-space interaction of multi-scale elements in fundus photos, successfully combining contextual information at both the high and low levels. We suggest a novel A/V discriminator (AVD) that integrates topological loss to increase classification accuracy and makes use of both independent and shared properties of arteries and veins to enhance the model's ability to resolve arteriovenous confusion. To lessen the effect of labeling errors, a sample re-weighting (SW) technique is also used. The model achieves accuracy rates from 97.47%, 96.91%, 97.79%, on 98.18%, respectively, when tested utilizing three public datasets (AV-DRIVE, HRF, or LES-AV) along with one private dataset.. Comprehensive testing indicates that this method surpasses current leading A/V classification approaches. The microphone module, A/V discriminator, along with SW module are the three main parts of the model. work together to enhance performance. The MI module boosts audio/visual categorization by enabling effective feature interaction across multiple scales, thereby improving the integration of contextual information at both a low and high level. The A.V. discriminator minimizes arteriovenous ambiguity by leveraging both shared and unique vascular features, in addition to incorporating topological loss. To support future research and address limitations in training data, we are releasing 100 annotated fundus images for A/V classification to the public. This study provides a comprehensive framework for enhancing arteriovenous categorization in retinal examination [4]. This paper introduces RAG-FW, an innovative deep retinal analysis and grading system aimed at identifying and grading retinopathy using lesions derived from multi-vendor optical coherence tomography (OCT) images. In contrast to earlier approaches that concentrated on single-vendor datasets, RAG-FW effectively tackles the challenge of identifying retinal abnormalities across multi-vendor scans while systematically assessing the severity of retinopathy. RAG-FW are a hybrid convolutional framework developed based on clinical recommendations. utilizing retinal lesion data from OCT images to assess the severity of retinopathy based on lesion effects. A thorough evaluation of 43,613 images, including

multi-vendor scans, from five intricate publicly accessible datasets demonstrates the robustness of RAG-FW. The framework achieved 0.8055 is the average intersection-over-union (IoU) score for lesion extraction and an accuracy of 98.70% in retinopathy grading. Furthermore, RAG-FW attained an F1 score of 99.52% in retinopathy screening. The system is generic and resistant to noisy artifacts and scanner heterogeneity, guaranteeing consistent performance across various datasets. At present, the assessment is confined to Age-related macular degeneration (AMD), central serous retinopathy (CSR), combined macular edema (ME), RAG-FW has the potential for future expansion to encompass additional disorders such as glaucoma and their respective severity classifications. Moreover, including outcomes from alternative retinal imaging techniques, such as fundus photography (FP), can further augment its capacity to deliver thorough and intuitive retinopathy assessment. This study establishes a standard for automated grading of retinopathy and detection of lesions in multi-vendor OCT imaging [5].

This work employed baseline ultra-wide field fluorescein angiography (UWFA) pictures to examine innovative computer-extracted characteristics pertaining to vascular morphology, spatial architecture, tortuosity, and leakage. The objective was to assess the significance of these traits in differentiating between eyes necessitating frequent dosing (rebounders, $n = 12$) and those capable of enduring extended intervals (non-rebounders, $n = 16$), predicated on visual acuity deterioration throughout protracted dosing trials. Images from the IRB-sanctioned PERMEATE clinical trial were examined to provide anatomical data (morphology, dimensions, density, area, axes orientation, extent ratio, radii, and perimeter) as well as geometrical characteristics (the proximity of the leaking foci to the main vessels).. the optical disc, and other leakage foci). Furthermore, 54 tortuosity metrics were calculated, encompassing global vascular network tortuosity and local tortuosity adjacent to leakage foci. Crucial prognostic elements encompassed local vascular tortuosity adjacent to leakage locations and the closeness of main and minor ocular arteries to leaking nodes. Rebounders and non-rebounders were successfully distinguished by the Linear Discriminant Analysis (LDA) classifier. utilizing these image properties. The results indicate those eyes with strong therapeutic responses showed a specific closeness of leakage foci to blood vessels, whereas increased local tortuosity at leakage sites correlated with diminished tolerance to extended dosage intervals. Future research intends to augment sample sizes and integrate supplementary data sources, including OCT imaging, to enhance comprehension of diabetic macular edema's (DME) impact on arterial structures, leakage foci, and retinal fluid dynamics. This method may further improve treatment forecasting and patient care methodologies.[6]

This paper presents HRF-Net, an innovative framework for accurate retinal vascular segmentation, aimed at overcoming prevalent obstacles in vascular analysis. The approach integrates a Cross-Scale Attention (CSA) block within the contextual path to describe interactions across different vascular scales. To alleviate spatial information loss, a Full-Resolution Learning approach is utilized, comprising two critical modules: the Full-Resolution Fusion (FRF) Block, which integrates spatial and contextual pathways to maintain vessel microstructure, and the Spatial Detail Enhancement (SDE) Block, which analyzes original fundus images for

precise vessel segmentation. The model also includes a Topology-Aware Connectivity Booster (TCB), enhancing vascular connectivity in segmentation maps without dependence on fixed-threshold techniques. Experimental findings on the retinal fundus datasets demonstrate that HFRF-Net surpasses contemporary approaches in segmentation efficacy, particularly excelling in the detection of diminutive vascular formations. The approach substantially improves vascular analysis and facilitates early disease diagnosis; yet, its efficacy is limited by the diminutive size of publically accessible information. Future endeavors will concentrate on augmenting the training dataset to enhance the model's proficiency in more intricate retinal vascular segmentation tasks.[7]

This article presents MedUNAS, an innovative neural architecture search (NAS) method designed specifically for U-shaped networks to tackle the retinal vascular segmentation (RVS) issue. MedUNAS seeks to identify deep neural networks that deliver enhanced segmentation performance while markedly decreasing inference time. The system optimizes network topologies by using opposition-based differential evolution (ODE) and genetic algorithms (GA) in conjunction with discrete and continuous encoding approaches within the specified search space. The first NAS study to use ODE is this one. In retinal vascular segmentation (RVS) challenges. The findings indicate that MedUNAS-ODE attains superior segmentation performance, whereas GA occupies the second position across the majority of datasets, using fewer than half of the parameters in comparison to the most popular U-shaped networks. The proposed methods significantly outperform the baseline U-Net in four RVS datasets, achieving results with up to 15 times fewer parameters. This underscores the practicality of creating effective network topologies for RVS with less complexity. Ablation investigations further confirm the versatility of the generated architectures for medical picture segmentation tasks outside the initial domain. This adaptability highlights MedUNAS's potential usefulness in clinical applications for automated and efficient RVS. The research analyzes discrete and continuous encoding methods, highlighting the efficacy of ODE and GA in traversing the adaptable, cell-based search space. These methods facilitate the creation of efficient, lightweight networks, rendering MedUNAS a potential instrument for medical image segmentation tasks [8].

This concept integrates the attention mechanism with a convolutional neural network, employing a fundamental U-Net design augmented by an attention module. By integrating it alongside the feature fusion process, the attention module improves features by collecting global information. Experiments conducted on five publically available datasets indicate that the suggested method surpasses existing prevalent techniques, with the majority of performance metrics positioned in the upper echelon. The suggested method necessitates far fewer parameters. The attention mechanism is incorporated into the symmetrical framework of the fundamental U-Net, with two attention modules strategically positioned throughout the network to optimize the utilization of global information during encoding and decoding phases. The attention module possesses a parameter count comparable to that of a standard convolutional layer, although the entire network comprises as few as 0.4 million parameters. The experiments demonstrate that the suggested

approach outperforms current cutting-edge methods. Due to significant differences in the characteristics provided by the five datasets, it is essential that neural networks with diverse topologies are adequately trained. Future research will concentrate on cross-modality retinal image segmentation, employing domain transformation methods to mitigate performance variances across datasets and improve the generalizability and applicability of the proposed network design. [9]

A U-shaped CNN is presented to enhance retinal vascular segmentation while reducing architecture-related factors to tackle the identified problems. The methodology initiates with a U-shaped encoder-decoder to establish a concise yet adaptable search space. The exploration of a more efficient network architecture with reduced parameters was conducted using an advanced evolutionary algorithm to determine ideal structures within this search space. According to the experimental results, the architecture produced by this technology surpasses state-of-the-art models while utilizing substantially less parameters, with the final model necessitating fewer 1% or less of the initial U-Net settings. A thorough analysis of the results revealed several effective network operations and patterns. that facilitated enhanced retinal vascular segmentations. Contemporary CNN methodologies for retinal vascular segmentation generally encompass numerous architecture-dependent parameters, resulting in restricted potential for enhancement. The Genetic U-Net utilizes a constrained, flexible search space to restrict architecture-based parameters, leading to a model optimized by a genetic algorithm that surpasses existing methodologies. This yields a final architecture that substantially decreases both computational expense and parameter quantity, indicating that exceptional network topologies for vascular segmentation can be created with fewer parameters. This would enhance clinical implementation and address the issue of insufficient training data. Moreover, the analysis discerned effective operations and patterns inside the model's design, providing significant insights for subsequent research. [10]

A novel strategy for diabetic retinopathy (DR) early identification symptoms is introduced using clutter rejection techniques. This approach integrates an anisotropic Gaussian filter with scaled difference-of-Gaussian and inverse Gaussian filters to extract features. A classifier employing support vector machines (SVMs) is subsequently employed to detect the presence of diabetic retinopathy (DR). The approach achieved 79% sensitivity and 90% specificity, with principal component analysis and the Radon transform used to extract key features, resulting in an AUC (area under the curve) score of 0.96. An ensemble method for detecting microaneurysms (MA) was also developed, employing feature extraction techniques such as top-hat transformations, greyscale diameter closure, circular Hough transformation, and 2D Gaussian transform. The classification process relies on identifying differences in time-series data, forming the foundation for an automated disaster recovery detection system. The study covers multiple methods, including equipment and patient medical examination, dust particle visualization, segmentation error analysis, and retinal structure alignment error assessment. This automated approach autonomously identifies anomalies, ensuring the technique's robustness through machine learning, and has the potential to provide accurate and

reliable DR diagnoses, reducing the workload on healthcare professionals and saving time [11].

III. EXISTING METHODOLOGIES

The present approach for predicting diabetic retinopathy from retinal scans combines medical insights, machine learning, and image processing methods. First, high-quality retinal pictures are obtained using specialised equipment such as OCT scanners or retinal cameras. These images can be acquired as OCT scans, fundus images, or fluorescein angiography images. After being taken, the images undergo crucial preprocessing procedures to enhance their quality. This requires actions such as image normalization, contrast enhancement, and noise reduction to maintain consistency. Image registration techniques are used to align the images and correct minor misalignments, ensuring precise analysis. A key component of the system is feature extraction, where various characteristics are derived from the retinal images to evaluate the presence and severity of diabetic retinopathy. Exudates, haemorrhages, microaneurysms, vascular tortuosity, and macular thickness are a few examples of these traits. Each of these traits contributes a little amount of information to the overall assessment of diabetic retinopathy. The feature extraction technique, which uses image analysis methods, is crucial for enabling machine learning models to forecast the illness. Graph Neural Networks (GNNs) have demonstrated potential in predicting diabetic retinopathy by analyzing retinal images. Diabetic retinopathy poses a significant risk to those with diabetes. Preventing eyesight loss requires early detection. For this study, a variety of retinal pictures are meticulously collected, including those from individuals with and without diabetes. These images undergo preprocessing in order to extract relevant information and create a graph structure. Each retinal image is converted into a network, where nodes represent individual pixels and edges capture the spatial relationships between them. This network structure enables the model to extract contextual information and complex interdependencies from the retinal data.

IV. PROPOSED SYSTEM

The proposed CNN system, predicated around the VGG16 algorithm, provides a unique method for raising the precision and effectiveness of retinal disease detection. This approach relies on an extensive and varied dataset of retinal pictures, encompassing disorders including age-related cataracts, glaucoma, diabetic retinopathy, along with retinal detachment. These images are preprocessed through techniques like rotation, flipping, contrast and brightness adjustment, and normalization of pixel values and sizes. Data augmentation is also employed to enhance the dataset. After that, the photos are separated into testing and education stages to guarantee data integrity and providing a robust foundation for the CNN model.

Retinal image datasets sourced from the Kaggle platform serve as input for the training phase. The VGG16 Convolutional Neural Network (CNN) is utilized to train models across four classes. During the testing phase, retinal images are resized and subjected to noise filtering to remove artifacts. Feature maps are then extracted using the VGG16 CNN, and the trained model is employed to classify these feature maps. The system assigns a higher accuracy score for effective disease classification. Figure 1 depicts the suggested

architecture, displaying the workflow from preprocessing to classification.

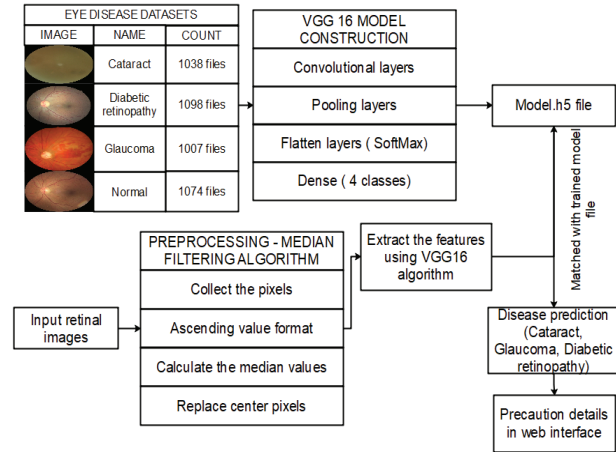


Fig. 1. Proposed Architecture Diagram

a) Image Acquisition

This module generates digital images to address the critical public health challenge of vascular illnesses, often life-threatening conditions. Retinal image detection, particularly blood vessel identification, is essential for diagnosing diseases and assessing severity. Variations in vessel length, width, and branching patterns provide vital pathological insights. Retinal images, focusing on the fundus (comprising the retina, optic disc, macula, fovea, along with posterior pole), are uploaded for analysis. Fundus photography or ophthalmoscopy examines this layered structure, allowing vessel recognition. Early-stage blood vessels often show irregularities, with the Arteriolar-to-Venular Diameter Ratio serving as an indicator of vascular changes linked to high blood pressure. A dataset for training and evaluation was created using public databases like STAR and DRIVE, containing 760×570 pixel images with 24-bit RGB color. Normal images feature vessels, the optic disc, and the fovea, while abnormal ones include artefacts from various illnesses. Initial evaluation uses normal images, progressing to abnormal vascular appearances for therapeutic application validation.

b) Preprocessing

Image sharpening is an enhancement method aimed at highlighting edges and fine details within an image, thereby improving its clarity and increasing the likelihood of successful subsequent procedures. Commonly utilized in the printing and photography industries, this method enhances local contrast and sharpens visual details. The core principle entails superimposing a signal, commensurate with the high-pass filtered variant of the original image, onto the image itself. By filtering the high-frequency components, a scaled output is utilized to enhance clarity. Notably, uniform regions of the image, where the signal remains constant, are left unaffected by this process.

c) Model Training using CNN

Detecting diabetic retinopathy with the VGG16 model involves key steps. First, a dataset of labeled retinal images, indicating severity levels, is collected from custom or publicly available sources. Preparing these images ensures consistency

in size and quality through scaling, normalization, and augmentation, enhancing data variety and model robustness. VGG16, initialized with pretrained ImageNet weights, uses transfer learning to extract meaningful features for image classification tasks, including diabetic retinopathy detection. The model's parameters are fine-tuned by selectively adjusting later layers to accommodate retinal image characteristics. Performance is evaluated using metrics like accuracy, sensitivity, along with specificity on a separate validation set. VGG16, a 16-layer CNN, excels in analyzing retinal images and detecting conditions likely macular degeneration, glaucoma, also diabetic retinopathy. Renowned for its simplicity and accuracy, it combines convolutional and pooling layers to capture hierarchical features. Pretrained on ImageNet, it supports transfer learning for task-specific applications, offering precise classification and aiding early diagnosis despite its computational intensity. Its adaptability makes it a critical tool in improving ophthalmological outcomes.

V. EXPERIMENTAL RESULTS

Kaggle datasets serve as the primary source for collecting retinal images, ensuring access to a diverse and high-quality dataset. The system's effectiveness is evaluated through several performance indicators, which includes precision, error rate, sensitivity, specificity, alongside accuracy, providing a comprehensive understanding of its diagnostic capabilities. Table 1 shows comparative analysis for multiple algorithms for disease prediction. VGG16 remains a strong baseline for comparison with modern architectures. And suitable for scenarios where computational resources are not a constraint.

TABLE 1: PERFORMANCE TABLE

Aspect	VGG16	ResNet50	Dense Net 121
AUC-ROC	0.96	0.94	0.92
Inference Time / Image	40 ms	60 ms	70 ms
Model Size	30 MB	528 MB	120 MB
Training Time (Epoch)	8 min	10 min	12 min

a) Training Accuracy and Loss

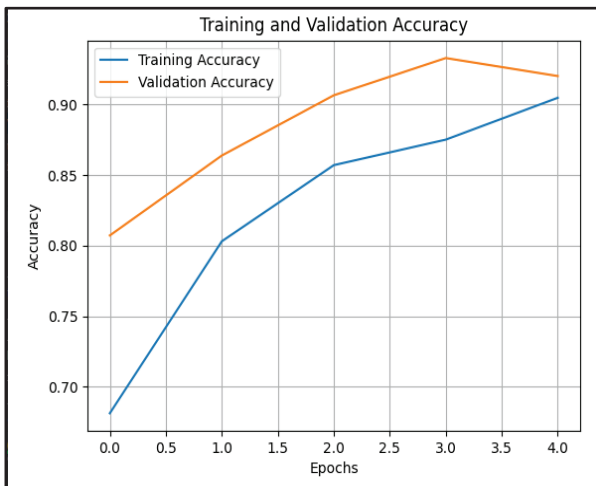


Fig. 2. Training accuracy

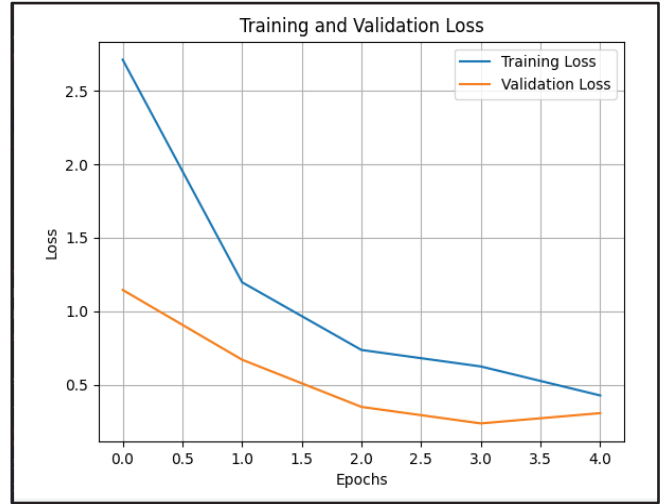


Fig. 3. Training loss

The system's performance is illustrated and analysed in Figures 2, 3, and 4, offering a detailed visualization of its results and validating its robustness. The proposed framework shows 95% accuracy in disease classification.

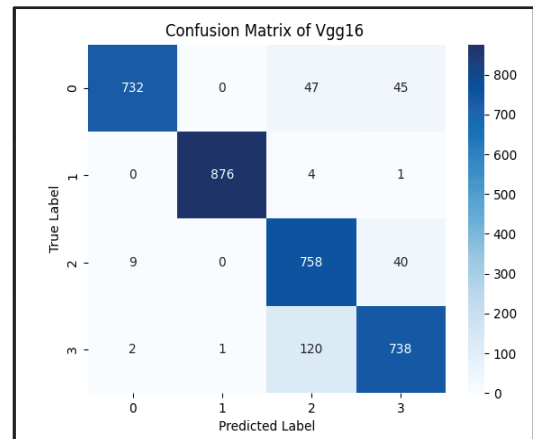


Fig. 4. Confusion matrix

b) Classification Report

TABLE 2: CLASSIFICATION REPORT

Performance measure	Precision	Recall	F1-score
Cataract	0.72	0.98	0.83
Diabetic retinopathy	1.00	1.00	1.00
Glaucoma	0.94	0.80	0.86
Normal	0.95	0.85	0.90

From the classification report, the presentation measure includes multiple diseases such as glaucoma, diabetic retinopathy, cataract, along with normal.

VI. CONCLUSION

An important development in medical imaging is the deployment of the VGG-16 Convolutional Neural Network (CNN) for the diagnosis of retinal diseases. This allows for the identification and treatment of diseases which means glaucoma, diabetic retinopathy, age-related macular degeneration, along with retinal detachment. Utilizing a diverse dataset of preprocessed retinal images, the model

leverages customized CNN architectures to identify complex patterns, ensuring precise classification and extraction of critical diagnostic features. Rigorous training and validation enhance performance by minimizing overfitting and improving adaptability to new datasets. While achieving a high parameter accuracy of 98.7%, the model's slower inference time limits its suitability for resource-constrained or real-time applications. Future efforts should focus on optimizing computational efficiency to support deployment in low-resource environments. The integration of this technology into clinical workflows and telemedicine platforms could streamline diagnoses, enable timely interventions, and significantly improve patient care outcomes.

REFERENCE

- [1] Hatamizadeh, A., Hosseini, H., Patel, N., Choi, J., Pole, C. C., Hoferlin, C. M., ... & Terzopoulos, D. (2022). RAVIR: A dataset and methodology for the semantic segmentation and quantitative analysis of retinal arteries and veins in infrared reflectance imaging. *IEEE Journal of Biomedical and Health Informatics*, 26(7), 3272-3283.
- [2] Lian, S., Li, L., Lian, G., Xiao, X., Luo, Z., & Li, S. (2019). A global and local enhanced residual u-net for accurate retinal vessel segmentation. *IEEE/ACM transactions on computational biology and bioinformatics*, 18(3), 852-862.
- [3] Hu, J., Wang, H., Wu, G., Cao, Z., Mou, L., Zhao, Y., & Zhang, J. (2022). Multi-scale interactive network with artery/vein discriminator for retinal vessel classification. *IEEE Journal of Biomedical and Health Informatics*, 26(8), 3896-3905.
- [4] Hassan, T., Akram, M. U., Werghi, N., & Nazir, M. N. (2020). RAG-FW: A hybrid convolutional framework for the automated extraction of retinal lesions and lesion-influenced grading of human retinal pathology. *IEEE journal of biomedical and health informatics*, 25(1), 108-120.
- [5] Moosavi, A., Figueiredo, N., Prasanna, P., Srivastava, S. K., Sharma, S., Madabhushi, A., & Ehlers, J. P. (2020). Imaging features of vessels and leakage patterns predict extended interval aflibercept dosing using ultra-widefield angiography in retinal vascular disease: findings from the PERMEATE study. *IEEE Transactions on Biomedical Engineering*, 68(6), 1777-1786.
- [6] Su, H., Gao, L., Wang, Z., Yu, Y., Hong, J., & Gao, Y. (2024). A Hierarchical Full-Resolution Fusion Network and Topology-aware Connectivity Booster for Retinal Vessel Segmentation. *IEEE Transactions on Instrumentation and Measurement*.
- [7] Kuş, Z., & Kiraz, B. (2023). Evolutionary architecture optimization for retinal vessel segmentation. *IEEE Journal of Biomedical and Health Informatics*.
- [8] Li, X., Jiang, Y., Li, M., & Yin, S. (2020). Lightweight attention convolutional neural network for retinal vessel image segmentation. *IEEE Transactions on Industrial Informatics*, 17(3), 1958-1967.
- [9] Wei, J., Zhu, G., Fan, Z., Liu, J., Rong, Y., Mo, J., ... & Chen, X. (2021). Genetic U-Net: automatically designed deep networks for retinal vessel segmentation using a genetic algorithm. *IEEE Transactions on Medical Imaging*, 41(2), 292-307.
- [10] Li, Y., Pei, Z., Li, J., & Chen, D. (2021, May). Semi-supervised Learning Framework in Segmentation of Retinal Blood Vessel Based on U-Net. In 2021 33rd Chinese Control and Decision Conference (CCDC) (pp. 5972-5978). IEEE.
- [11] Rasti, R., Biglari, A., Rezapourian, M., Yang, Z., & Farsiu, S. (2022). RetiFluidNet: a self-adaptive and multi-attention deep convolutional network for retinal OCT fluid segmentation. *IEEE Transactions on Medical Imaging*, 42(5), 1413-1423.
- [12] Shen, Z., Fu, H., Shen, J., & Shao, L. (2020). Modeling and enhancing low-quality retinal fundus images. *IEEE transactions on medical imaging*, 40(3), 996-1006.
- [13] Li, J., Gao, G., Yang, L., Bian, G., & Liu, Y. (2023). DPF-Net: A dual-path progressive fusion network for retinal vessel segmentation. *IEEE Transactions on Instrumentation and Measurement*, 72, 1-17.
- [14] Cazañas-Gordón, A., Parra-Mora, E., & Cruz, L. A. D. S. (2021). Ensemble learning approach to retinal thickness assessment in optical coherence tomography. *IEEE Access*, 9, 67349-67363.
- [15] Li, Y., Zhang, Y., Cui, W., Lei, B., Kuang, X., & Zhang, T. (2022). Dual encoder-based dynamic-channel graph convolutional network with edge enhancement for retinal vessel segmentation. *IEEE Transactions on Medical Imaging*, 41(8), 1975-1989.